IBM BOB HACKATHON · TRACK 2 — MODERNIZE WHAT MATTERS

ParityProof

Modernize COBOL to Java and **prove the new code behaves exactly like the original** — a copybook-aware runner diffs the COBOL reference and the Java twin field-by-field, on identical inputs.

✔ It runs

Two original COBOL batch programs compiled under GnuCOBOL; Java 21 twins; edge-case input generators. One command: `bash run.sh`.

✔ It proves

100% field parity (64/64 fields), and it catches real divergence (injected rounding bug → 83% on ACCOUNT-INT, exact fields pinpointed).

✔ It is central to IBM Bob

Bob explains the rules (Ask mode), plans a change, and edits the Java (Agent mode); a parity hook re-runs the proof after Bob's edits (one real firing is logged). Sessions and reviews are committed.

Team: Abdul Qaadir (Java lead · Bob driver) · Satish (Cloud · CI) · Karthik (Business analysis · story)

THE PROBLEM

Banks want to modernize. They're afraid to.

80%+ of transactions

Run on IBM i / IBM Z, much of it in COBOL batch that has run untouched for decades.

Senior COBOL developers are retiring

The pattern is a knowledge cliff — the people who understand the business rules are leaving.

Modernization risk is the blocker

"Tests pass" on generated code means nothing until you've shown the new output matches the old.

The question every bank asks first is not "did you generate code?" — it's "is it the same?"

WHY THIS WINS

Many teams work with Bob. Few prove the output is correct.

The first thing any bank wants is a guarantee the rewrite is behaviorally identical. We built the mechanism that delivers it.

Most teams

Show generated Java → judge sees code, must trust it.

ParityProof

Adds **a running proof the Java matches the COBOL byte-for-byte**, with Bob in the loop (Ask / Agent modes).

Judging edge

Correctness + explainability is the defensible innovation, not the generation itself.

100%
field parity — 64/64 (2 programs)

83%
→ FAIL on ACCOUNT-INT when a rounding slip is injected (20/24) — fields pinpointed

14
edge-case records: 6 interest + 8 transaction posting

WHAT WE BUILT

ParityProof pipeline

- **COBOL reference runtime** — GnuCOBOL (verified 3.2.0) runs two original batch programs: ACCOUNT-INT and TXN-POST (signed amounts, COMP-3). A Dockerfile exists but is not yet built.
- **Business-rule catalog** — drafted with Bob and Claude Code, line citations verified; Karthik's business review is pending.
- **Java 21 twins** — plain single-file programs; BigDecimal arithmetic; copybook layout → fixed-width output.
- **Edge-case input generators** — half-cent rounding, zero and tiny balances, negative balances, sign handling, and field-limit overflow (silent COBOL truncation, reproduced).
- **Parity runner** — diffs every field of every record, per program, by the copybook layout → result.json; exits non-zero on any mismatch.
- **Gate + dashboard** — `run.sh` fails unless parity is 100%; a GitHub Actions workflow is written (not yet run on GitHub); a static dashboard is built (hosting planned).

Every output keeps the original fixed-width format so the diff is exact.

LIVE PROOF

Parity catches real bugs

Interest batch, acct 1: COBOL `00000043208` (432.08). Java with rounding changed to DOWN (injected): `00000043207` (432.07).

CORRECT
HALF_UP → 100% pass, 64/64 fields

INJECTED BUG
DOWN → ACCOUNT-INT 20/24 FAIL, INTEREST + NEW-BAL flagged (a deliberate mutation, not a Bob change)

Every run is saved under **evidence/runs/<timestamp>/** (inputs, both outputs, result.json): a real PASS and a real injected FAIL are committed, and the runner also fails cleanly when a COBOL or Java output is missing.

HOW IBM BOB IS CENTRAL

Bob in the loop at every step

Ask — extract rules (skill used)

Bob explained TXN-POST as business rules with source-line citations — all citations verified. Review caught its mistakes and kept its real finding (batch total repeated on every record; no per-account grouping).

Plan a change (in Ask mode)

Bob planned a comment-only change to the Java twin, citing the COBOL lines, with a parity run as the acceptance gate.

Agent — edit the Java

Bob made a comment-only edit with `apply_diff`; the **parity hook fired automatically** and logged **64/64 PASS** (evidence/hooks/). A FAIL firing is not yet captured.

Governance

Rules file (COMP-3 → BigDecimal, COBOL truncation, cite source lines) + cobol-explainer skill + parity hook script.

Adoption evidence

Bob replies, our reviews of them and six screenshots (skill used, hooks active, edit) are committed under **bob/evidence-files/**. The Bobalytics export is still to be added.

TECH STACK

Built with

GnuCOBOL 3.2 | Java 21 / BigDecimal | Single-file Java, no build step | Bash pipeline (run.sh) | Static HTML dashboard | GitHub Actions (written) | Dockerfile (not yet built) | IBM Bob (Ask / Agent modes)

Correctness infrastructure

Copybook-driven field diff; BigDecimal arithmetic; signed and COMP-3 fields; fixed-width I/O so diffs are exact.

Repeatability

One command (`bash run.sh`) reproduces the whole run and keeps timestamped evidence. Docker and CI definitions are provided but not yet exercised.

Evidence machine-readable

result.json drives the static dashboard: overall and per-program parity, per-field rates, mismatch drill-down.

FUTURE ROADMAP & LEARNINGS

Next steps

- **Business review** — Karthik signs off the rule catalogs; decide whether the silent overflow truncation (preserved for parity) should become a guard in production.
- **True packed-decimal files** — our files are DISPLAY text; binary COMP-3 record I/O is the next hard case.
- **Ship the gate** — run the CI workflow on GitHub; host the dashboard.
- **More programs** — statements and other batch jobs; AWS CardDemo source once its license is confirmed.

Key learning

Parity-as-a-test-harness changes how a migration is trusted: it turns an article of faith ("Bob's code works") into a measurable, verifiable guarantee — which is exactly the evidence a bank asks for before signing off. It also surfaced hidden COBOL behavior (silent overflow truncation) that a naive rewrite would have "fixed" and broken parity.